The cost of preparing a network to learn

A sharp norm formula for zero-output deep linear networks
and an isotropic-to-specialized adaptation transition

An autonomous mathematical investigation

31 October 2026

Abstract

A network can initially predict zero while being prepared to learn different directions at different rates. We determine the exact minimum squared parameter norm needed to prescribe that initial learning matrix in a scalar-output deep linear network of arbitrary fixed depth. For a desired matrix $G \succeq 0$ on $\mathbb{R}^d$, the minimum at depth $L \geq 2$ is $(L-1)[\operatorname{tr} G-(d-1)\lambda_{\min}(G)]^{1/(L-1)}$. We also obtain sharp hidden-width and parameter-count formulas. An attaining construction implements the desired preconditioned update exactly in one finite simultaneous gradient step. The resulting feasible set yields an explicit statistical optimum: with increasing norm budget, a learner for a known rank-r task family passes from isotropic plasticity to mixed plasticity to full specialization. Its minimum first hidden width rises from 1 to $r+1$, then falls to r. All statements are proved and supported by reproducible numerical checks. The exact extremal formulas were not located in the inspected literature; historical originality remains uncertain.

Status. These are proved statements within the specified model, independently audited and numerically checked. They have not been externally peer reviewed or verified in a formal proof assistant. They are not claims about biological brains, nonlinear foundation models, or the cost of discovering a useful prior. Historical priority is not established.

Question. The motivating distinction is between inheriting an answer and inheriting a mechanism that makes later learning effective. We isolate it by requiring the initial predictor to be identically zero and prescribing its sensitivity to parameter learning. Squared parameter norm is the resource proxy studied below; it is not literal biological energy.

1 A precise model of preparation for learning

Consider a depth-L, scalar-output linear network

$$f_\theta(x) = W_L W_{L-1} \cdots W_1 x, \qquad x \in \mathbb{R}^d, \quad L \geq 2.$$

The hidden widths $m_1, \ldots, m_{L-1}$ are positive integers. Every matrix entry is an independent trainable parameter. There are no biases, tied weights, residual connections, or width-dependent normalization factors. Both the parameter norm and gradient descent use these fixed Euclidean coordinates. Define

$$E(\theta) = \sum_{\ell=1}^{L} \|W_\ell\|_F^2.$$

We impose the initial condition $W_L \cdots W_1 = 0$: the network initially predicts zero for every input, while its derivatives need not vanish.

Its neural tangent kernel, the Gram kernel of parameter derivatives [1], has the form

$$\langle \nabla_\theta f_\theta(x), \nabla_\theta f_\theta(y) \rangle = x^\top G_\theta y, \qquad G_\theta \succeq 0.$$

The question is: what is the least parameter norm required to prescribe the entire matrix G_θ, while retaining the zero predictor?

2 Exact norm, width, and parameter costs

For $G \succeq 0$, write

$$\lambda = \lambda_{\min}(G), \qquad \Gamma(G) = \operatorname{tr} G - (d-1)\lambda.$$

Equivalently, if $\mu_1, \ldots, \mu_d$ are the eigenvalues,

$$\Gamma(G) = \lambda + \sum_{i=1}^{d} (\mu_i - \lambda).$$

Thus $\Gamma(G) \geq 0$, with equality exactly when $G = 0$.

Theorem 1 (Exact cost of a prescribed tangent matrix)**.** *For every $G \succeq 0$, the minimum of $E(\theta)$ over depth-L networks satisfying $f_\theta \equiv 0$ and $G_\theta = G$, allowing compatible positive hidden widths, is*

$$\boxed{E_{\min,L}(G) = (L-1)\Gamma(G)^{1/(L-1)}.}$$

For $G \neq 0$, let

$$m_* = \operatorname{rank}(G - \lambda I_d) + \mathbf{1}_{\{\lambda > 0\}}.$$

A prescribed architecture admits such an initialization if and only if $m_1 \geq m_*$. Every such architecture attains the same minimum norm. The minimum number of trainable scalar parameters over these architectures is

$$\boxed{P_{\min,L}(G) = (d+1)m_* + (L-2).}$$

The minimum parameter count and minimum norm are attained simultaneously.

For $G = 0$, all-zero weights attain zero norm in every architecture. The width formula above is restricted to $G \neq 0$, since positive hidden widths exclude width zero.

2.1 The universal lower bound

Define forward and backward products by

$$R_\ell = W_\ell \cdots W_1, \quad R_0 = I_d, \qquad b_\ell^\top = W_L \cdots W_{\ell+1}, \quad b_L = 1,$$

and put $q_\ell = \|b_\ell\|_2^2$. Differentiating a single layer gives

$$\nabla_{W_\ell} f_\theta(x) = b_\ell (R_{\ell-1} x)^\top.$$

Consequently,

$$G_\theta = \sum_{\ell=1}^{L} q_\ell R_{\ell-1}^\top R_{\ell-1}. \tag{1}$$

The first summand is $q_1 I_d$, hence $\lambda_{\min}(G_\theta) \geq q_1$ and

$$\Gamma(G_\theta) \leq q_1 + \sum_{\ell=2}^{L} q_\ell \|R_{\ell-1}\|_F^2 =: S. \tag{2}$$

Set $p_0 = 1$ *artificially*; this is not the squared Frobenius norm of $R_0 = I_d$. For $\ell \geq 1$, set $p_\ell = \|R_\ell\|_F^2$. The null-product condition gives $p_L = q_0 = 0$, and $q_L = 1$. Define

$$\alpha_\ell = \begin{cases} p_\ell / p_{\ell-1}, & p_{\ell-1} > 0, \\ 0, & p_{\ell-1} = 0, \end{cases} \qquad \beta_\ell = \begin{cases} q_{\ell-1}/q_\ell, & q_\ell > 0, \\ 0, & q_\ell = 0. \end{cases}$$

Vanishing prefixes and suffixes remain zero, so $p_\ell = \alpha_\ell p_{\ell-1}$ and $q_{\ell-1} = \beta_\ell q_\ell$ hold without exceptions. In particular, $\alpha_L = \beta_1 = 0$.

Lemma 2. *For every layer,* $\|W_\ell\|_F^2 \geq \alpha_\ell + \beta_\ell$.

Proof. At layer one, $\alpha_1 = \|W_1\|_F^2$ and $\beta_1 = 0$. For $\ell \geq 2$, first suppose $p_{\ell-1} > 0$ and $q_\ell > 0$. Decompose

$$W_\ell = \frac{b_\ell b_\ell^\top W_\ell}{q_\ell} + A_\ell.$$

The summands are Frobenius-orthogonal. The first has squared norm $\|W_\ell^\top b_\ell\|_2^2 / q_\ell = \beta_\ell$. Moreover, $b_\ell^\top R_\ell = 0$, so this summand annihilates $R_{\ell-1}$, and $A_\ell R_{\ell-1} = R_\ell$. Submultiplicativity therefore gives

$$\|A_\ell\|_F^2 \geq \frac{\|R_\ell\|_F^2}{\|R_{\ell-1}\|_F^2} = \alpha_\ell.$$

If $q_\ell = 0$, then $\beta_\ell = 0$ and the forward norm inequality alone suffices. If $p_{\ell-1} = 0$, then $\alpha_\ell = 0$ and the backward norm inequality suffices. If both vanish, the claim is immediate. □

The recurrences express S as

$$S = \sum_{\ell=1}^{L} \left(\prod_{j<\ell} \alpha_j \right) \left(\prod_{j>\ell} \beta_j \right).$$

These L terms are among the nonnegative terms obtained by expanding $\prod_{j=1}^{L-1} (\alpha_j + \beta_{j+1})$. Applying AM–GM and the lemma,

$$\begin{aligned} \Gamma(G_\theta) \leq S &\leq \prod_{j=1}^{L-1} (\alpha_j + \beta_{j+1}) \\ &\leq \left(\frac{\sum_j \alpha_j + \sum_j \beta_j}{L-1} \right)^{L-1} \leq \left(\frac{E(\theta)}{L-1} \right)^{L-1}. \end{aligned}$$

Rearranging proves the norm lower bound.

2.2 A construction attaining the bound

Assume $G \neq 0$, and put

$$c = \Gamma(G)^{1/(L-1)}, \qquad t = c^{L-2}.$$

Construct $W \in \mathbb{R}^{m_* \times d}$ and $a \in \mathbb{R}^{m_*}$ such that

$$W^\top W = (G - \lambda I_d)/t, \qquad \|a\|_2^2 = \lambda/t, \qquad W^\top a = 0. \tag{3}$$

An eigendecomposition supplies the rows of W associated with the positive eigenvalues of $G - \lambda I_d$. If $\lambda > 0$, append one zero row and place a, with length $\sqrt{\lambda/t}$, on this extra coordinate. If $\lambda = 0$, take $a = 0$.

The two-layer network $a^\top W x$ is identically zero. Its tangent matrix is $W^\top W + \|a\|_2^2 I_d = G/t$, and its energy is

$$\|W\|_F^2 + \|a\|_2^2 = \Gamma(G)/t = c.$$

Append $L-2$ scalar layers, each equal to $\sqrt{c}$. Their parameter derivatives initially vanish because the two-layer predictor vanishes. Their product multiplies the earlier derivatives by $c^{(L-2)/2}$, so the full tangent matrix is exactly G. Total energy is $(L-1)c$. Padding with zero rows and columns embeds this construction in any architecture with first width at least m_* and all other widths positive, without changing the kernel or energy.

2.3 Why the width and parameter count are also minimal

Equation (1) implies

$$G_\theta = q_1 I_d + W_1^\top H W_1 \quad \text{for some } H \succeq 0.$$

If $m_1 < d$, the matrix W_1 has a nontrivial nullspace. Thus $\lambda_{\min}(G_\theta) = q_1$. If $q_1 > 0$, the null-product condition $W_1^\top b_1 = 0$ also gives $\operatorname{rank} W_1 \leq m_1 - 1$. Accordingly,

$$\operatorname{rank}(G_\theta - q_1 I_d) \leq m_1 - \mathbf{1}_{\{q_1 > 0\}},$$

which proves $m_1 \geq m_*$. If $m_1 \geq d$, this inequality is automatic since $m_* \leq d$.

The first weight matrix therefore has at least dm_* entries, the second at least m_*, and each of the remaining $L-2$ matrices at least one. Widths $(m_*, 1, \ldots, 1)$ attain the resulting count. This finishes the proof of Theorem 1.

3 A finite first step, not merely a tangent approximation

Corollary 3. *The optimal construction above realizes the update*

$$w_{\text{new}} = -\eta G \nabla \mathcal{L}(0)$$

exactly after one simultaneous gradient-descent step of any step size η, for every differentiable loss $\mathcal{L}(w)$ on its effective linear predictor w.

Proof. Write $w = pW^\top a$, where p is the scalar suffix product, and let $g = \nabla \mathcal{L}(0)$. Initially the suffix gradients vanish, while

$$a' = a - \eta p W g, \qquad W' = W - \eta p\, a g^\top.$$

Hence

$$\begin{aligned} pW'^{\top}a' &= -\eta p^2(W^{\top}W + \|a\|_2^2 I_d)g + \eta^2 p^3 g(a^{\top}Wg) \\ &= -\eta G g, \end{aligned}$$

using $W^{\top}a = 0$. The zero-kernel case is immediate. □

The two-factor gradient expansion underlying this cancellation is already documented in [5]. This corollary is asserted for the explicit construction; it does not assert exact linear behavior for arbitrary deep null networks or for subsequent training steps.

4 Exact prescription can cost more than domination

The cost is not monotone in the positive-semidefinite order when $d \geq 3$. For example, in dimension three,

$$G = \operatorname{diag}(4, 3, 0), \qquad H = \operatorname{diag}(4, 3, 3)$$

satisfy $H \succeq G$, but

$$\Gamma(G) = 7, \qquad \Gamma(H) = 4.$$

Prescribing unequal directional sensitivities can therefore cost more than allowing additional sensitivity.

More generally,

Proposition 4. *For every $G \succeq 0$,*

$$\min_{H \succeq G} E_{\min,L}(H) = (L-1)\lambda_{\max}(G)^{1/(L-1)}.$$

Proof. For every $H \succeq 0$, $\Gamma(H) \geq \lambda_{\max}(H)$. If $H \succeq G$, this is at least $\lambda_{\max}(G)$. Equality is attained by $H = \lambda_{\max}(G) I_d$, whose Γ value is $\lambda_{\max}(G)$. □

This is a resource comparison between metrics. A larger tangent matrix does not by itself guarantee better statistical performance or stable training at a fixed finite learning rate.

5 Rigidity of generic optimal constructions

The equality conditions also restrict optimal architecture.

Proposition 5. *Suppose* $\operatorname{rank}(G - \lambda_{\min}(G) I_d) \geq 2$. *Every norm-minimizing realization satisfies* $W_2 W_1 = 0$ *and* $\operatorname{rank} W_\ell \leq 1$ *for* $\ell \geq 2$. *For* $\ell \geq 3$, *it additionally satisfies* $\|W_\ell\|_F^2 = \Gamma(G)^{1/(L-1)}$.

Proof. Equality in (2) forces $q_1 = \lambda_{\min}(G)$; the rank hypothesis then implies $\operatorname{rank} W_1 \geq 2$. If $R_2 \neq 0$, the strict inequality $\|W_1\|_{\mathrm{op}} < \|W_1\|_F$ makes the layer-two forward bound strict, contradicting equality in the cost bound. Thus $R_2 = 0$. Since $G \neq 0$, equation (1) now forces $q_2 > 0$. All later forward prefixes vanish. Equality in the backward projection bounds makes every W_ℓ, $\ell \geq 2$, have rank at most one. Finally, equality in AM–GM makes $\alpha_1 + \beta_2, \beta_3, \ldots, \beta_L$ all equal to $\Gamma(G)^{1/(L-1)}$. For $\ell \geq 3$, the backward bound is an equality and $\|W_\ell\|_F^2 = \beta_\ell$, proving the assertion. □

6 A resource-driven transition from general plasticity to specialization

The metric-cost formula has a statistical consequence: a small initialization norm can make isotropic plasticity optimal even when every task belongs to a known proper subspace. The conclusion concerns the following precisely specified class of linear first-step learners.

Fix integers $2 \le r < d$, $n \ge 1$, and a rank-r orthogonal projection $P \in \mathbb{R}^{d\times d}$. Independently for each task, draw a vector θ with

$$\mathbb{E}\theta = 0, \qquad \mathbb{E}[\theta\theta^\top] = S = P/r.$$

Draw independent inputs $x_i \sim N(0, I_d)$, independently of θ, and set

$$y_i = x_i^\top \theta + \xi_i, \qquad z = \frac{1}{n}\sum_{i=1}^{n} x_i y_i.$$

The noises are mutually independent, independent of the task and design, and have mean zero and variance $\sigma^2 \ge 0$. Neither the task distribution nor the noise distribution needs to be Gaussian. We choose a deterministic $G \succeq 0$ before seeing the new task and estimate θ by $\widehat{\theta} = Gz$. The loss is $\mathbb{E}\left\|\widehat{\theta} - \theta\right\|^2$; it also equals expected squared prediction error on an independent isotropic test input, excluding test-label noise.

For depth $L \ge 2$ and squared initialization norm budget B, Theorem 1 identifies the feasible metrics as

$$\Gamma(G) \le k, \qquad k = \left(\frac{B}{L-1}\right)^{L-1}, \qquad \Gamma(G) = \operatorname{tr} G - (d-1)\lambda_{\min}(G). \tag{4}$$

The gradient step size is fixed at one. The resulting optimization is global over this PSD linear-update class. It does not optimize arbitrary nonlinear finite updates of every deep network. The cost-minimizing null-network constructions nevertheless realize these linear updates exactly in their first finite gradient step.

6.1 Exact risk and the three regimes

Write

$$c = \frac{n+1}{n}, \quad \rho = \frac{1+\sigma^2}{n}, \quad \alpha = c + r\rho, \quad \beta = (d-r)\rho.$$

Gaussian fourth moments give

$$\mathbb{E}[z\theta^\top] = S, \qquad \mathbb{E}[zz^\top] = cS + \rho I_d.$$

Indeed, $\mathbb{E}[xx^\top Sxx^\top] = 2S + \operatorname{tr}(S)I_d = 2S + I_d$; the off-diagonal sample pairs contribute S, and the label noise contributes $\sigma^2 I_d/n$. Thus

$$R(G) = 1 - 2\operatorname{tr}(GS) + c\operatorname{tr}(G^2S) + \rho\operatorname{tr}(G^2). \tag{5}$$

Theorem 6 (Exact specialization transition)**.** *Define*

$$k_0 = \frac{r-1}{\alpha(r-1) + \beta r}, \qquad k_1 = \frac{r}{\alpha}, \qquad D = \alpha(r-1)^2 + \beta r^2.$$

The unique minimizer of (5) *under* $G \succeq 0$ *and* (4) *is*

$$G_k = a(k)P + b(k)(I_d - P),$$

where

$$(a(k), b(k)) = \begin{cases} (k, k), & 0 \le k \le k_0, \\ \left(\dfrac{(r-1)^2 + \beta r k}{D}, \dfrac{(r-1)(r - \alpha k)}{D} \right), & k_0 < k < k_1, \\ (1/\alpha, 0), & k \ge k_1. \end{cases} \tag{6}$$

Its risk is

$$R_*(k) = \begin{cases} 1 - 2k + (\alpha + \beta)k^2, & 0 \le k \le k_0, \\ 1 - \dfrac{1}{\alpha} + \dfrac{\beta(r - \alpha k)^2}{\alpha D}, & k_0 < k < k_1, \\ 1 - 1/\alpha, & k \ge k_1. \end{cases} \tag{7}$$

The minimum first hidden width needed to realize this optimal metric is

$$m_{\min}(G_k) = \begin{cases} 1, & 0 < k \le k_0, \\ r + 1, & k_0 < k < k_1, \\ r, & k \ge k_1. \end{cases}$$

Proof. The feasible set is convex because $-\lambda_{\min}$ is convex, and compact because $\Gamma(G) \ge \lambda_{\max}(G)$ for $G \succeq 0$. The objective is strictly convex: its quadratic part contains $\rho \|G\|_F^2$ with $\rho > 0$, and the remaining quadratic term is convex.

Average a feasible metric over orthogonal transformations preserving range P and its orthogonal complement. Convexity preserves feasibility and cannot increase the invariant objective. Uniqueness therefore forces $G = aP + b(I_d - P)$ for $a, b \ge 0$. Equation (5) becomes

$$R(a, b) = 1 - 2a + \alpha a^2 + \beta b^2. \tag{8}$$

If $b > a$, reducing b to a lowers both the risk and the cost $(d - r)b - (d - r - 1)a$. Hence $a \ge b$, and the budget becomes

$$ra - (r - 1)b \le k.$$

For fixed a, the best choice is consequently

$$b = \max\left\{0, \frac{ra - k}{r - 1}\right\}, \qquad 0 \le a \le k.$$

The unrestricted optimum $(a, b) = (1/\alpha, 0)$ is feasible exactly when $k \ge k_1$. If $0 < k < k_1$, the risk decreases throughout $0 \le a \le k/r$, including its right derivative at k/r. The optimum therefore has $a > k/r$ and $b = (ra - k)/(r - 1)$.

The unconstrained minimizer of this last quadratic is

$$\widetilde{a} = \frac{(r-1)^2 + \beta r k}{D} > k/r.$$

It satisfies $\widetilde{a} \ge k$ precisely when $k \le k_0$. Clipping it to the upper endpoint $a = k$ proves (6); direct substitution proves (7). The case $k = 0$ is immediate.

Finally, the width formula accompanying Theorem 1 is

$$m_{\min}(G) = \operatorname{rank}(G - \lambda_{\min}(G) I_d) + \mathbf{1}_{\{\lambda_{\min}(G) > 0\}}.$$

The three regimes have respectively $a = b > 0$, $a > b > 0$, and $a > b = 0$, which gives the claimed widths. □

The positive b in the middle regime is statistically costly: it contributes βb^2 despite the absence of task signal outside the subspace. But it reduces the initialization cost $ra - (r-1)b$ while preserving a. At small budgets this economy dominates, making the optimum completely isotropic. The metric varies continuously across both thresholds, although its exact minimum width follows the nonmonotone sequence $1 \to r+1 \to r$.

6.2 A concrete instance

For $d = 6$, $r = 2$, $n = 4$, and $\sigma^2 = 1$,

$$\alpha = \frac{9}{4}, \qquad \beta = 2, \qquad k_0 = \frac{4}{25}, \qquad k_1 = \frac{8}{9}.$$

Between the thresholds,

$$a(k) = \frac{4 + 16k}{41}, \qquad b(k) = \frac{8 - 9k}{41}.$$

At $k = 0.4$, the optimum has $a \approx 0.253659$, $b \approx 0.107317$, and risk 0.660488. Requiring $b = 0$ instead gives $a = 0.2$ and risk 0.69. The optimum thus benefits from retaining plasticity in four directions that contain no source-task signal. The corresponding minimum widths are 1, 3, and 2 across the three regimes.

6.3 The isotropic threshold for an arbitrary task spectrum

The initial isotropic regime is not specific to equal eigenvalues within a subspace. Keep the same observations and let the task second moment be any $S \succeq 0$ with $\operatorname{tr} S = 1$.

Proposition 7 (Exact isotropic threshold)**.** *The unique optimum is $G = kI_d$ if and only if*

$$0 \le k \le k_{\text{iso}} := \frac{1 - \lambda_{\max}(S)}{c(1 - \lambda_{\max}(S)) + (d-1)\rho}.$$

If S is nonscalar, the optimum is nonisotropic for every $k > k_{\text{iso}}$. In particular, a positive isotropic interval exists whenever $\operatorname{rank} S > 1$.

Proof. At $G = kI_d$, the risk gradient is $A = 2[(ck - 1)S + \rho k I_d]$. The translation identity uses the same spectral formula for Γ on all symmetric matrices. In particular, $\Gamma(kI_d + H) = k + \Gamma(H)$ identifies feasible directions by $\Gamma(H) \le 0$. Write $H = tI_d + Z$ with $Z \succeq 0$ and $t = \lambda_{\min}(H)$. Feasibility gives $t = -\operatorname{tr} Z - s$ for some $s \ge 0$, so

$$\operatorname{tr}(AH) = \operatorname{tr}\big((A - \operatorname{tr}(A) I_d) Z\big) - s \operatorname{tr} A.$$

This is nonnegative for all feasible directions exactly when $\operatorname{tr} A \le 0$ and $A - \operatorname{tr}(A) I_d \succeq 0$; necessity follows by taking $H = -I_d$ and then rank-one Z with $s = 0$. The trace condition implies $ck < 1$.

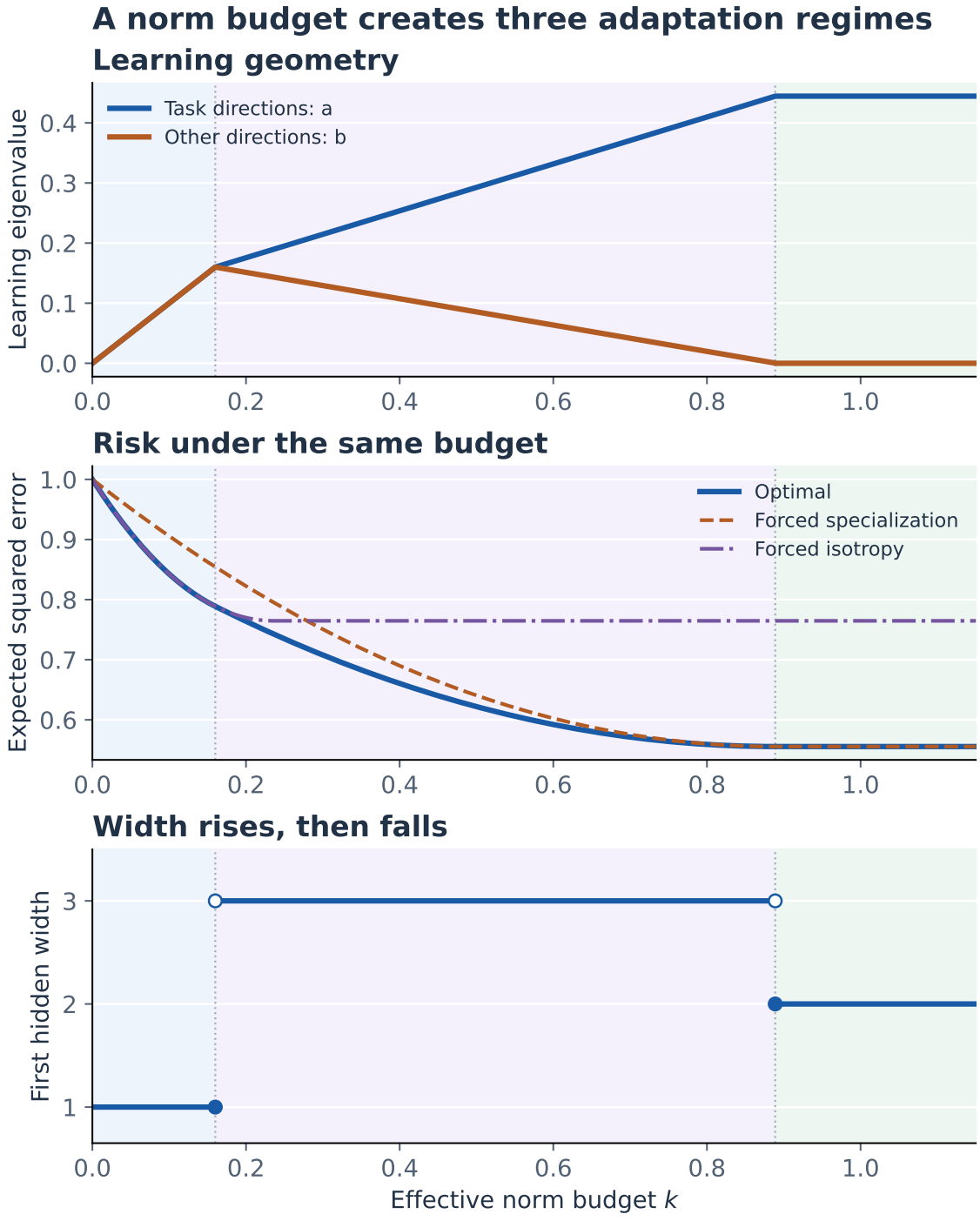


Figure 1: The exact resource-dependent optimum for the example above. The first transition introduces task specialization while retaining positive plasticity outside the task subspace; the second eliminates that remaining plasticity. The horizontal coordinate is the effective metric budget k, related to the squared parameter norm budget by (4).

The smallest eigenvalue of the second matrix then corresponds to $\lambda_{\max}(S)$ and gives the displayed threshold, which itself implies the trace condition since $\lambda_{\max}(S) \geq 1/d$. Convexity makes these conditions sufficient, and for $k > 0$ positive definiteness of kI_d makes them locally necessary. The case $k = 0$ is immediate.

An isotropic optimum aI_d with $0 < a < k$ would be an interior feasible point and would require $(ca-1)S+\rho aI_d = 0$, impossible for nonscalar S. The boundary $a = k$ has already been characterized, while $a = 0$ is suboptimal when $k > 0$. This proves the remaining claims. □

7 Computational checks and a finite-step limitation

The arguments above are proofs; the computations in this section are independent checks of formulas, constructions, and implementations. Finite experiments cannot establish a universal inequality or historical originality. All numbers below were read from the saved outputs `results_plasticity.json`, `audit_geometry_results.json`, and `results_phase.json`. The accompanying scripts record their random seeds and use double-precision arithmetic.

7.1 Energy, kernel realization, and derivatives

Two implementations used different null-network generators. The first projected a randomly selected layer off its backward sensitivity vector (or zeroed the output layer when selected); the audit chose the final row in the left nullspace of the preceding matrix product. The first tested 5,600 networks at depths 2–8 and input dimensions 1–8; the independent audit tested 12,600 networks at depths 2–8 and input dimensions 1–6. Neither found a violation beyond floating-point roundoff. The largest reported normalized bound ratio in each implementation was 1.0000000000000002. These ratios use different, equivalent normalizations of the proved inequality and should not be treated as identical statistics.

Check	Cases	Largest discrepancy
Random null networks, both implementations	18,200	Roundoff only
Attaining constructions: energy	3,500	5.33×10^{-15}
Attaining constructions: prescribed kernel	3,500	6.29×10^{-15}
Attaining constructions: exact one-step response	3,500	8.00×10^{-15}
Independent complex-step kernel check	42	1.43×10^{-14}
First hidden width in the independent audit	2,100	0

The entries for the constructions take the larger error reported by the two implementations. The independent audit includes positive definite, singular, repeated-minimum-eigenvalue, isotropic, and zero kernels. Its null-function residual is exactly zero for all attaining constructions; for its randomly generated networks the largest absolute residual is 4.36×10^{-15}. The other implementation reports a largest relative null residual of 5.77×10^{-16}; absolute and relative residuals are deliberately not pooled.

To avoid checking the analytic kernel formula only against another copy of that formula, the audit also differentiates the actual matrix product by the complex-step method. Across 42 networks it computes 5,341 individual parameter derivatives, assembles the resulting Jacobian, and compares its Gram matrix with the analytic kernel. This check is independent of the prefix–suffix formula used

by the main implementation. All listed discrepancies are consistent with floating-point evaluation, but that observation is not a formal rounding-error bound.

7.2 The specialization transition

The phase-transition experiment uses $d = 6$, task-subspace dimension $r = 2$, $n = 4$ examples per task, and noise variance $\sigma^2 = 1$. The predicted transition budgets are 0.16 and 8/9. A common sample of 200,000 simulated tasks evaluates six budgets; consequently the estimates at different budgets are correlated. The numerical values below provide a direct check of the closed-form expected risk.

Budget k	a	b	Analytic risk	Simulated risk
0.08	0.080000	0.080000	0.867200	0.867523
0.16	0.160000	0.160000	0.788800	0.789256
0.40	0.253659	0.107317	0.660488	0.661089
0.80	0.409756	0.019512	0.559024	0.560121
8/9	0.444444	0	0.555556	0.556806
1.20	0.444444	0	0.555556	0.556806

Every simulation error is less than one estimated standard error. A separate diagonal quadratic program, with linear constraints, succeeds at all six budgets and agrees with the predicted eigenvalues to within 3.80×10^{-15}. Its smallest recorded constraint slack is -1.29×10^{-14}, a numerical rather than exact feasibility statement.

We also retain unsuccessful diagnostics. Direct optimization over a randomly rotated full matrix encounters nonsmooth minimum-eigenvalue constraints: four of six SLSQP runs do not report success. The largest constraint violation is approximately 2.73×10^{-7}, and one matrix differs from the analytic answer by about 3.00×10^{-4} in Frobenius norm. Those runs are neither optimality certificates nor counterexamples. They remain in the saved output; the successful diagonal quadratic programs and the analytical proof are the relevant checks.

7.3 Zero output alone does not make a finite step linear

The exact finite-step response is a property of the attaining construction, not of every null network. For a rational counterexample, take one input, depth three, and

$$W_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad W_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad W_3 = \begin{pmatrix} 0 & 1 \end{pmatrix}.$$

The effective coefficient $w = W_3 W_2 W_1$ is zero, the scalar kernel is $G = 3$, and the energy is $E = 5$. Consider the squared loss $\frac{1}{2}(w - u)^2$, corresponding to input $x = 1$ and target u. Put $h = \eta u$. One simultaneous gradient-descent step gives

$$W_1' = \begin{pmatrix} 1 \\ h \end{pmatrix}, \qquad W_2' = \begin{pmatrix} 1 & 1 \\ h & 1 \end{pmatrix}, \qquad W_3' = \begin{pmatrix} h & 1 \end{pmatrix}.$$

Direct multiplication yields

$$w' = 3h + h^2 = \eta G u + \eta^2 u^2.$$

Thus the kernel still gives the first-order response, while a quadratic term survives at finite step size. This example does not attain the energy bound: its energy 5 exceeds $2\sqrt{3}$. It illustrates why the special construction, and the cancellation checked in its proof, are essential to the finite-step claim.

8 Originality assessment

Several ingredients are established mathematics or established learning concepts:

- Parameter-gradient kernels describe learning geometry [1]; deep linear factorization can act as a preconditioner [2].
- Meta-learning can learn a geometry for subsequent adaptation [3].
- Antisymmetric or mirrored initializations can make the initial predictor identically zero [4].
- The exact two-factor gradient expansion is known [5]. Its cancellation at a zero product is not claimed as a standalone discovery.
- Minimum weight norm for a prescribed end-to-end linear map is a studied representation-cost problem [6].
- Minimum sharpness of deep linear realizations with a prescribed end-to-end map has been studied [7, 8].
- NTK isotropy has recently been connected directly to preserving plasticity [9].
- Developmental programs with spontaneous pre-experience activity already exist as an AI research direction [10].

The specific candidate contributions are the sharp cost

$$(L-1)[\operatorname{tr} G-(d-1)\lambda_{\min}(G)]^{1/(L-1)},$$

the joint width and parameter-count characterization, and the resulting norm-constrained specialization and width transitions. The general task-spectrum onset threshold is a further consequence. These prescribe both a zero predictor and its full tangent matrix while minimizing the initialization norm. This differs from minimizing norm for a prescribed output, or minimizing sharpness with the output fixed.

Searches used general and exact terminology, including *NTK representation cost*, *minimum norm prescribed tangent kernel*, *deep linear zero function*, *zero-output initialization*, *deep linear Jacobian norm*, and *isotropic specialization norm budget*. Both search engines available in the research environment were used, and primary texts below were inspected at their relevant definitions or results. No equivalent exact cost formula or transition theorem was located.

This is non-exhaustive evidence. Equivalent results could appear under matrix-product derivative optimization, factorization sensitivity, or another name. The exact formula's absence from these searches does not certify novelty. This note separates a proof of the mathematics from an assessment of historical originality.

Three other directions were explored and deprioritized. Diffusion-history rank led toward block-Krylov controllability and majorization. Instantaneous spontaneous-activity graph identification led toward graph cancellation ambiguities. Robust task-covariance design led toward classical minimax linear filtering. Those branches helped select a narrower candidate; no additional priority claims are made for them.

9 Significance and limitations

The main theorem gives an exact cost for preparing a learning response while holding the current predictor fixed at zero. It provides a constructive answer, an obstruction that applies to every width, and a matching lower bound on the number of parameters.

Its spectral cost separates a shared isotropic component from excess directional sensitivity. This makes some increases in the learning matrix reduce its realization cost. In the statistical application, a cheap isotropic component competes with noise from irrelevant directions, producing two exact resource thresholds and a nonmonotone width requirement. This is a substantive consequence of the architecture's parameterization.

The following boundaries are essential.

1. **Model scope.** The cost theorem assumes scalar-output, bias-free, fully connected deep linear networks, with independent trainable entries and a fixed Euclidean parameter metric. Nonlinear activations, vector outputs, tied weights, residual paths, and normalization can change the result.
2. **Resource definition.** Squared weight norm does not count the information encoded in G or its eigenspaces, evolutionary search, meta-training, development time, or numerical precision. The separate parameter-count theorem is still not a genome-description-length theorem. Zero-valued parameters have zero norm cost but still exist and can learn.
3. **Learning-rate normalization.** The phase transition fixes the learning rate to one. An unbounded free learning rate can trade off against initialization scale and invalidate a norm-only cost interpretation. A different fixed rate can be incorporated by rescaling the effective matrix and budget.
4. **One-step scope.** The attaining construction has an exact finite first step. Subsequent kernels generally change. The statistical optimum is among positive semidefinite linear first-step maps, not every nonlinear estimator or arbitrary finite deep-network update.
5. **Known task prior.** The task projector is assumed available. The theorem does not account for learning or encoding it, nor prove robustness when the inherited projector is wrong.
6. **Widths are realization requirements.** The sequence $1, r+1, r$ describes the minimum first hidden width for the optimal matrices. It does not prove that an evolving network will spontaneously follow this sequence.
7. **Numerical and historical limits.** Finite experiments can expose errors but cannot establish a theorem. The proofs here provide the mathematical argument. The literature investigation does not establish historical priority.

Further questions include a vector-output analogue with an operator-valued kernel; cost formulas with tied parameters or a fixed topology; a joint norm-plus-parameter-count budget; the effect of anisotropic or non-Gaussian inputs; and exact two-step adaptation. These are research directions, not established extensions.

Final verdict. This investigation produced a proved extremal theorem and proved statistical consequences. Independent mathematical checks and executed numerical experiments support the derivations. The exact formulas appear distinct from the inspected sources, but their historical novelty remains unresolved.

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